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SEA4001W

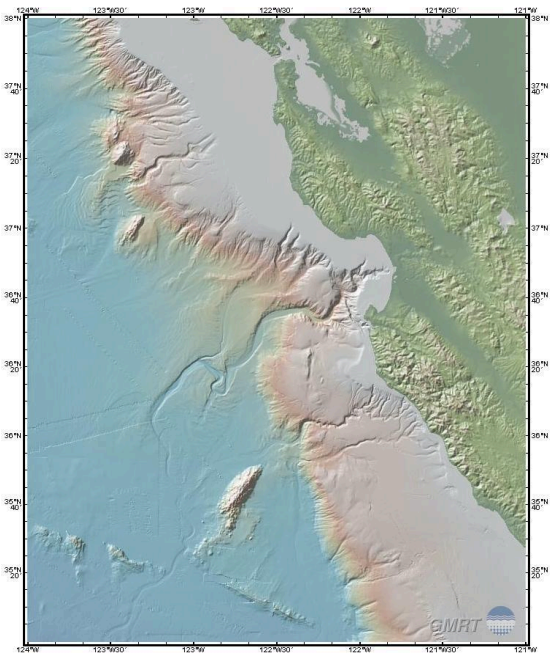
Department of Oceanography

Honours - 20/3/2026

P4 - Final Assignment: Putting It All Together

Exploring The Relationship Between Chlorophyll-*a* Concentration and Bathymetry In Monterey Bay, California.

Marcello Vichi



Ryan *et al.* (2009)

Introduction

Global marine productivity is disproportionately concentrated within the four Eastern Boundary Upwelling Systems (EBUS) (Bograd *et al*., 2022). Monterey Bay is located at 36.8°N and 121.8°W within the California Current System. It is a region characterised by intriguing bathymetric features and strong fluctuations in chlorophyll-*a* concentrations, especially across seasons (Manzer *et al*., 2019).

The bay has a pronounced morphological submarine canyon feature, with the canyon-head beginning close to the coastline (Styles, Wakamatsu and Fischer, 2025). Also, the area is well researched for its extensive marine life and complex food webs driven by upwelling. Positioned on the eastern boundary of the North Pacific Subtropical Gyre, the bay's inner shelf and canyon are primarily composed of warm waters surrounded by cooler ocean waters (Pennington *et al*., 2018).

According to Manzer *et al*. (2019), the circulation patterns within the bay are a mesoscale cyclonic flow during the upwelling season in the early spring. During this time, cold, upwelled water reportedly enters the gyre from the South in Point Sur, with an anticyclonic flow outward of the mouth.

To determine whether phytoplankton activity is predominantly controlled by wind-driven upwelling or by light availability, Monterey Bay represents a significant area to explore the changes in chlorophyll-*a* concentration both spatially and temporally.

Data and Methods

Bathymetry Data

The bathymetry data were obtained from the Global Multi-Resolution Topography (GMRT) Synthesis platform. In the file format of Coards/CF Compliant NetCDF (GMT id: nd). At an unmasked setting with the grid resolution of medium, 244.60 m/node. The file size is roughly 9MB, with a grid width of 1366 nodes and a grid height of 1700 nodes. The projection used is WGS84 Lat/Lon (EPSG:4326), alongside the GMRT version 4.4, made available in January 2026. With one data variable 'altitude' displayed under the Cylindrical Equidistant projection (Ryan *et al.*, 2009)

Extent boundaries:

West: -124.0°

- East: -120.9°
- South: 34.9°
- North: 38.0°

Chlorophyll Data

The Chlorophyll dataset named (ESA CCI (Climate Change Initiative) Ocean Colour Product, Version 5.0) is a three-dimensional NetCDF file consisting of data which begins from September 1997 to January 2020. Its extent covers the globe, creating a high-resolution 4 km grid representing linear concentration of chlorophyll derived from a blended algorithm approach (OCI, OCI2, OC2, and OCx), classified as a Level-3. It was produced by Plymouth Marine Laboratory (PML) for the European Space Agency using SeaWiFS, MODIS, and VIIRS sensors (Sathyendranath *et al*., 2020).

- Latitude (4320 values)
- Longitude (8640 values)
- Chlorophyll-a concentration (chloro_a -mg m^{-3}. not log-transformed)
- Start date = '1997-12-01', End date = '1998-08-01' *for assignment*

Results

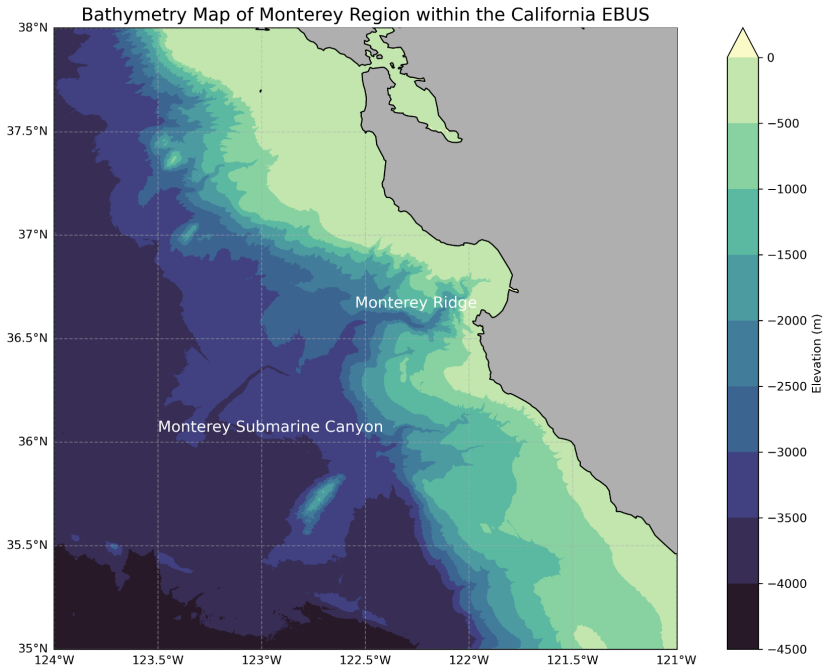


Figure 1: GEBCO bathymetry map of the Monterey Region within the Cal-EBUS.

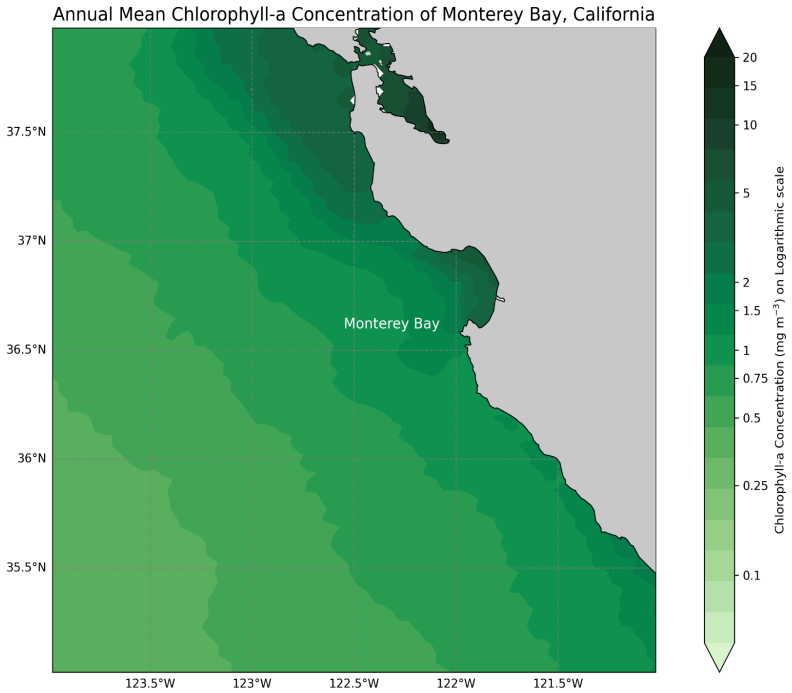


Figure 2: Annual Mean Chl-*a* Concentration (on logarithmic scale) of Monterey Bay Region in California. Chlorophyll is highest on the continental shelf near Monterey, decreasing with increasing distance away from the shoreline.

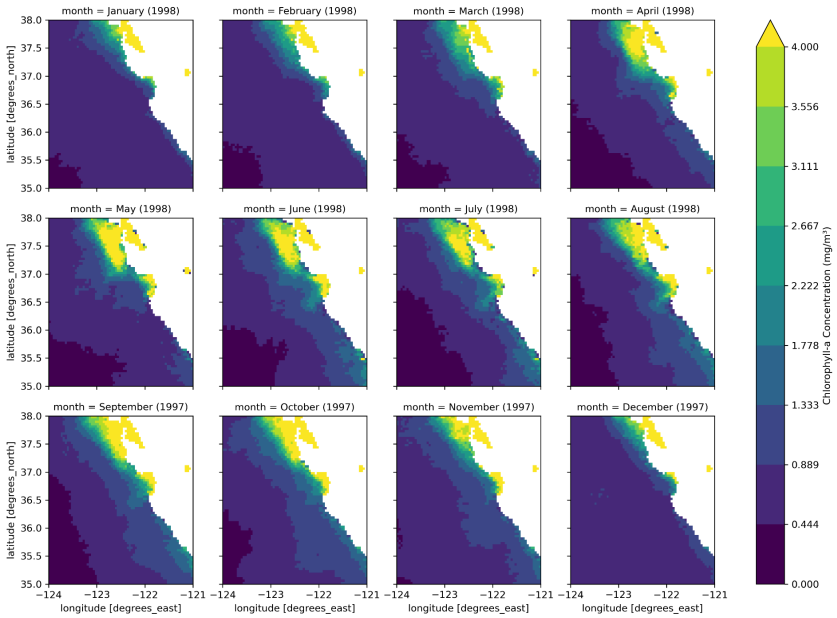


Figure 3: Monthly Climatologies of Chl-a Concentration across 1997 to 1998 (months).

Upwelling intensities within the region increased during June and declined during the July-August period with lessened northwesterly winds. Dominant months of phytoplankton bloom are consistent with Manzer *et al*. (2019) findings, with maximum concentration within the bay northwards in June (summer season).

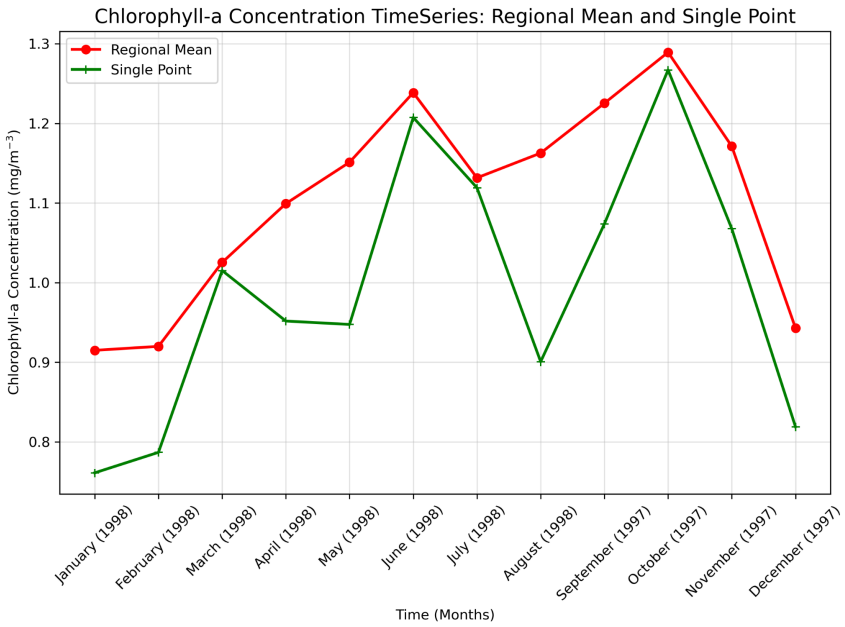


Figure 4: Chlorophyll-a Concentration (mg/m^{-3}) across (1997-1998) in months.

The single point is a selected centralised point within the bay region and follows a similar pattern to the regional mean, with the strongest deviation in August 1998.

References

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